

CA_eSUN v1.0 — Combination Reachability

San Diego Sheriff live configuration, hand built by engineering. Generated 2026-09-02 13:50 from the emitted JSON, the devdoc Basic Queries list, and the metadata XML. Every verdict below is computed, not asserted.

Headline: 8 of 25 built combinations are UNREACHABLE.

- 17 reachable / 8 unreachable of 25 built combinations.
- 16 combinations carry a POISONED condition array. Any value-comparison operator (IN / NOT_EQUALS / NOT_IN) disables the *entire* conditions array — live-proven platform behaviour — so the combination fires *unconditioned* and first-match hands every fill to the earliest matching combination.
- 12 combinations have no test in the generated plan, which is the same set: a combination that can never win produces no test.
- 2 combination(s) are not authorised by the devdoc at all.

Method: each combination's own set[] is filled and the canonical firing walk (Get-FiringKeyRef, the same predicate the simulator and log-attribution gates use) is asked what actually wins — in-state and out-of-state. If a different keyRef wins both times, an officer cannot reach that combination and the wire carries the winner instead.

Article

Query	keyRef	Status	Fires instead	Plan tests	Devdoc	set[]	Conditions
ArticleSingleQuery	ArticleSerialNumber	REACHABLE		5	Basic	PurposeCode, ArticleSerialNumber	ArticleTypeCode+ArticleBrand EXCLUSIVE

Boat

Query	keyRef	Status	Fires instead	Plan tests	Devdoc	set[]	Conditions
BoatQuery	PurposeCodeBoatHullIdNumber	REACHABLE		2	Basic	PurposeCode, BoatHullIdNumber	(none)
BoatQuery	PurposeCodeRegistrationNumber	REACHABLE		2	Basic	PurposeCode, RegistrationNumber	(none)

Firearm

Query	keyRef	Status	Fires instead	Plan tests	Devdoc	set[]	Conditions
GunQuery	PurposeCodeFirearmSerialNumber	REACHABLE		5	Basic	PurposeCode, SerialNumber	(none)
GunQuery	PurposeCodeNameLastAge	REACHABLE		1	Basic	PurposeCode, NameFirst, NameLast, Age	BirthDate+Age EXCLUSIVE
GunQuery	PurposeCodeNameLastBirthDate	REACHABLE		1	Basic	PurposeCode, NameFirst, NameLast, BirthDate	BirthDate+Age EXCLUSIVE

Person

Query	keyRef	Status	Fires instead	Plan tests	Devdoc	set[]	Conditions
AFS	PurposeCodeNameLastAge	REACHABLE		0	NOT IN DEVDOC	PurposeCode, NameFirst, NameLast, Age	BirthDate+Age EXCLUSIVE; RegistrationState IN [POISONED]
AFS	PurposeCodeNameLastBirthDate	REACHABLE		0	NOT IN DEVDOC	PurposeCode, NameFirst, NameLast, BirthDate	BirthDate+Age EXCLUSIVE
DriverHistoryQuery	KQName	UNREACHABLE	L1Name	0	Basic	PurposeCode, NameFirst, NameLast, SexCode, BirthDate, RegistrationState	RegistrationState NOT_EQUALS [POISONED]
DriverHistoryQuery	KQOperatorLicenseNumberIn	REACHABLE		0	Basic	PurposeCode, OperatorLicenseNumber	RegistrationState IN [POISONED]
DriverHistoryQuery	KQOperatorLicenseNumberOut	UNREACHABLE	KQOperatorLicenseNumberIn	0	Basic	PurposeCode, OperatorLicenseNumber, RegistrationState	RegistrationState NOT_EQUALS [POISONED]
DriverHistoryQuery	L1Name	REACHABLE		0	Basic	PurposeCode, NameFirst, NameLast	RegistrationState IN [POISONED]
DriverLicenseQuery	DQNameBirthSexCodeOut	UNREACHABLE	L1NameIn	0	Basic	PurposeCode, NameFirst, NameLast, BirthDate, SexCode, RegistrationState	RegistrationState NOT_EQUALS [POISONED]
DriverLicenseQuery	KQOperatorLicenseNumberOut	UNREACHABLE	L1OperatorLicenseNumberIn	0	Basic	PurposeCode, OperatorLicenseNumber, RegistrationState	RegistrationState NOT_EQUALS [POISONED]
DriverLicenseQuery	L1NameIn	REACHABLE		2	Basic	PurposeCode, NameFirst, NameLast	RegistrationState IN [POISONED]
DriverLicenseQuery	L1OperatorLicenseNumberIn	REACHABLE		2	Basic	PurposeCode, OperatorLicenseNumber	RegistrationState IN [POISONED]
DriverLicenseQuery	VPAGEIn	UNREACHABLE	L1NameIn	0	Basic	PurposeCode, NameFirst, NameLast, Age	BirthDate+Age EXCLUSIVE; RegistrationState IN [POISONED]

DriverLicenseQuery	VPBirthDateIn	UNREACHABLE	L1NameIn	0	Basic	PurposeCode, NameFirst, NameLast, BirthDate	BirthDate+Age EXCLUSIVE; RegistrationState IN [POISONED]
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Vehicle

Query	keyRef	Status	Fires instead	Plan tests	Devdoc	set[]	Conditions
CAIlojackQuery	LojackID	REACHABLE		1	Transactions Supported (NOT Basic)	PurposeCode, LojackID	(none)
VehicleRegistrationQuery	4LicensePlateTypeIn	UNREACHABLE	QVLicensePlateNumberIn	0	Basic	PurposeCode, LicensePlateNumber, LicensePlateTypeCode	RegistrationState IN [POISONED]
VehicleRegistrationQuery	IdentificationNumberInOut	REACHABLE		5	Basic	PurposeCode, VehicleIdentificationNumber	(none)
VehicleRegistrationQuery	NameAddressIn	REACHABLE		5	Basic	PurposeCode, NameFirst, NameLast	RegistrationState IN [POISONED]
VehicleRegistrationQuery	QVLicensePlateNumberIn	REACHABLE		2	Basic	PurposeCode, LicensePlateNumber	RegistrationState IN [POISONED]
VehicleRegistrationQuery	RQLicensePlateTypeYearOut	UNREACHABLE	QVLicensePlateNumberIn	0	Basic	PurposeCode, LicensePlateNumber, RegistrationState	RegistrationState NOT_IN [POISONED]
VehicleRegistrationQuery	VPNameBirthDateIn	REACHABLE		2	Basic	PurposeCode, NameFirst, NameLast, BirthDate	RegistrationState IN [POISONED]

Devdoc — Basic Queries Supported (the authority for WHICH queries may exist)

ArticleSingleQuery &middledot; BoatQuery &middledot; DriverHistoryQuery &middledot; DriverLicenseQuery &middledot; GunQuery &middledot; VehicleRegistrationQuery

- **AFS** — built with 2 combinations, and it appears **nowhere in the devdoc**. Not Basic, not under Transactions Supported.
- **CAIlojackQuery** — built with 1 combination. It *is* in the devdoc, but under **Transactions Supported**, not Basic Queries Supported. Its devdoc combination is 1. (In/Out) LojackIdentifier, [OriginatingAgencyORI, RelatedHitSearchIndicator].

What this does not say

- An UNREACHABLE verdict is about **routing**, not about whether the query works. The winning combination still sends a valid request — it simply is not the one the configuration appears to intend.
- Reachability here is computed from the JSON. It has **not** been confirmed against a live wire capture; no tenant test package exists for v1.0 yet.
- The separate validator findings (7 FAIL, of which 6 are the AP #1 / AP #2 codeTypeProvider regression introduced between the 2026-08-17 and 2026-09-02 exports) are recorded in the provider BUILD_NOTES and are not repeated here.